

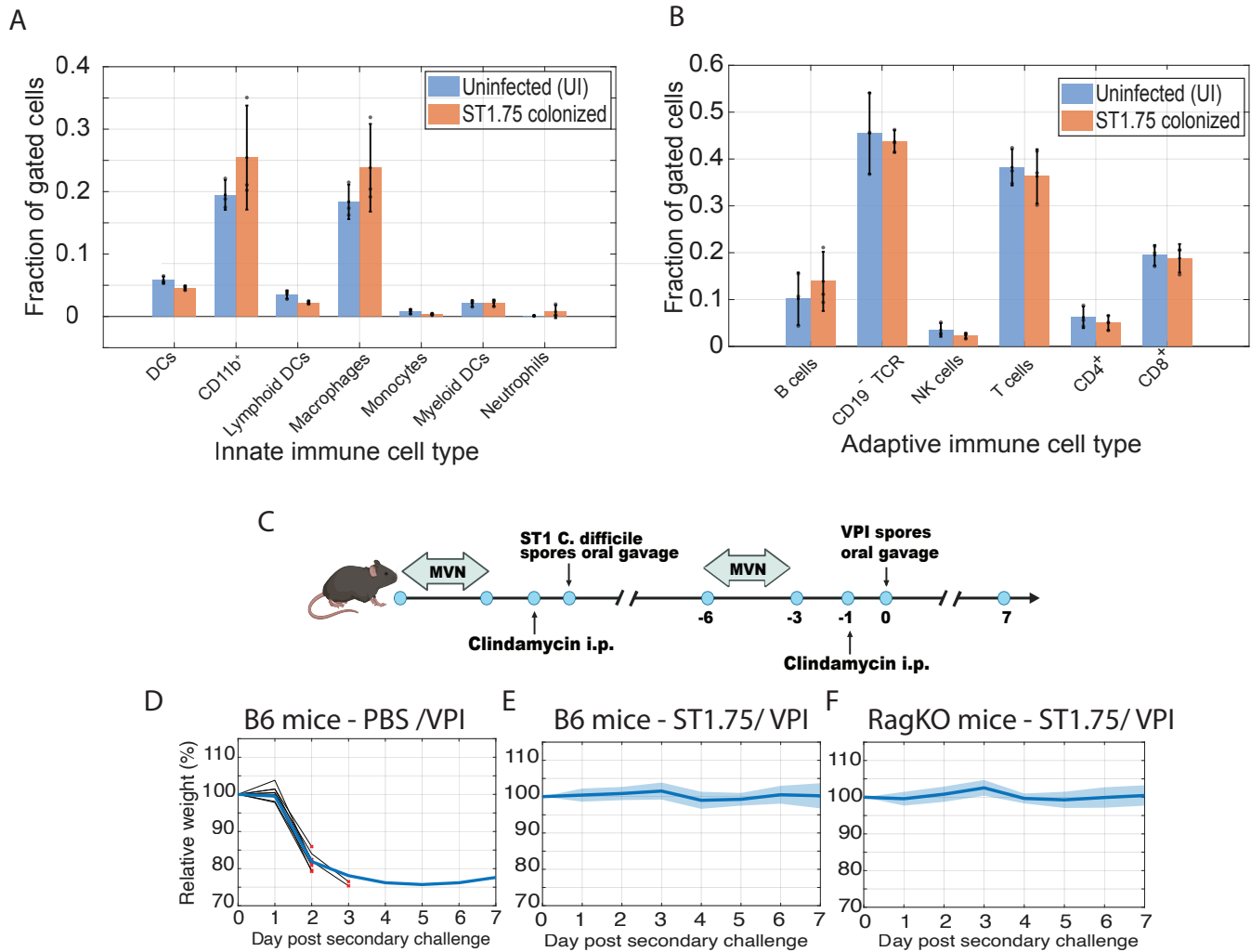


Figure 2: Protection conferred by ST1.75 is independent of adaptive immunity. (A) Fraction of gated cells for innate immune cell populations in the colon and mesenteric lymph nodes of uninfected (UI, blue) and ST1.75-colonized (orange) mice, quantified by multi-parameter flow cytometry. Cell types shown include DCs, CD19-TCR β - cells, NK cells, CD11b⁺ cells, macrophages, monocytes, and neutrophils. No significant differences were observed between groups (Wilcoxon rank-sum, $n = 3$ per group, all $p \geq 0.1$). Error bars indicate SEM. (B) Fraction of gated cells for adaptive immune cell populations in the colon and mesenteric lymph nodes of uninfected (UI, blue) and ST1.75-colonized (orange) mice. Cell types shown include B cells, CD4⁺T cells, CD8⁺ T cells, T cells, $\gamma\delta$ T cells, and DCs. No significant differences were observed between groups (Wilcoxon rank-sum, $n = 3$ per group, all $p \geq 0.1$). Error bars indicate SEM. (C) Schematic of the three-arm experiment. C57BL/6 mice received either PBS (sham) or ST1.75 spores as the primary infection, and RAG1^{-/-} mice received ST1.75 as the primary infection. All groups were subsequently rechallenged with VPI10463 spores following the same MVN antibiotic and clindamycin regimen. (D-F) Relative weight trajectories following VPI10463 secondary challenge for (D) PBS-pretreated B6 mice (sham), (E) ST1.75-pretreated B6 mice, and (F) ST1.75-pretreated RAG1^{-/-} mice ($n = 10$ per group). Shaded regions indicate 95% confidence intervals.